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Geophysical Studies of North African Cenozoic Volcanic Areas: II. Jebel Soda, Libya

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This study is the second of a series in which a main concern will be to try to establish the Neogene absolute motion of the African Plate. A combined paleomagnetic and K–Ar whole rock age dating study has been made of the Jebel Soda basaltic volcanic area of central Libya. Sixty basaltic flows were sampled in an area centered on 28°45′N, 15°30′E. Precisely defined paleomagnetic directions were obtained for 57 of the flows. Considerable attention has been given to the problem of data selection to obtain a best measure of the pole representing the dipole field during the period of volcanic activity. The equatorial nature of the directions, with equivalent poles extending continuously to 30°N latitude, provide a problem in data selection. Similar characteristics are apparent in the results of Schult and Soffel (1973) for the same area. A provisional average pole for the combined data sets, located at 73°N, 195°E (K: 26, A_{95}: 5.6°) is significantly different from the geographic pole. Reasons for the 17° divergence of paleomagnetic and geographic poles are discussed briefly.

Ceci est la deuxième étude faisant partie d'une série d'études visant d'essayer essentiellement d'établir le mouvement absolu de la Plaque Africaine au Néogène. Une étude combinée, paléomagnétique et datation de l'âge de la roche au K–Ar a été faite sur la région volcanique basaltique de Jebel Soda, Lybie Centrale. Soixante coulées basaltiques ont été échantillonnées dans une région localisée à 28°45′N, 15°30′E. Des directions paléomagnétiques précises ont été obtenues pour 57 échantillons de ces coulées. Une attention considérable a été donnée au problème de la sélection des résultats afin d'obtenir la meilleure mesure du pôle représentant le champ bipolaire pendant la période d'activité volcanique. La nature équatoriale des directions, avec des pôles équivalents s'étendant continuellement à une latitude de 30°N présente un problème pour la sélection des données. Des caractéristiques similaires apparaissent aussi dans les résultats de Schult et Soffel (1973) sur la même région. Un pôle provisoire moyen pour les groupes de résultats combinés localisé à 73°N, 195°E (K: 26, A_{95}: 5.6°) est significativement différent du pôle géographique. Les raisons pour la divergence de 17° des pôles paléomagnétiques et géographiques sont discutées brièvement. [Traduit par le journal]

Introduction

This is the second in a series of papers in which precisely defined paleomagnetic pole positions and potassium argon whole-rock ages are given for North African Cenozoic volcanic areas (Fig. 1). Paper I describes results for the Haruj Assuad area (Ade-Hall *et al.* 1974). The purpose of these investigations is to provide data for estimates of the absolute motion of the African Plate during this crucial period. Currently there is disagreement as to whether this plate has been essentially stationary during the Neogene (Burke

and Wilson 1972) or has been in motion with typical major-plate velocity (Morgan 1972). This paper gives data for the Jebel Soda basaltic area, a region centered on 29.5°N, 15.5°E. The Jebel Soda basalts cover an area of approximately 6000 km^2, and in contrast with the adjacent Haruj Assuad basalts, while still generally quite fresh are deeply eroded and characterized by an absence of preserved volcanic landforms. These features testify to the relative age of the Jebel Soda basalts. Eruption during the Miocene has been suggested (reported in Klitzsch 1970). Schult and Soffel (1973) have recently published paleomagnetic results for the Jebel Soda area.

Sampling

Sampling was carried out using portable diamond drills of the U.S.G.S. type. Sixty flows were sampled in two vertical sequences, with four separate geographically or solar oriented drill cores being taken from each flow. Both sections were located close to the highway from Socna in the Jofra group of oases to Sebha in the Fezzan (Table 1).

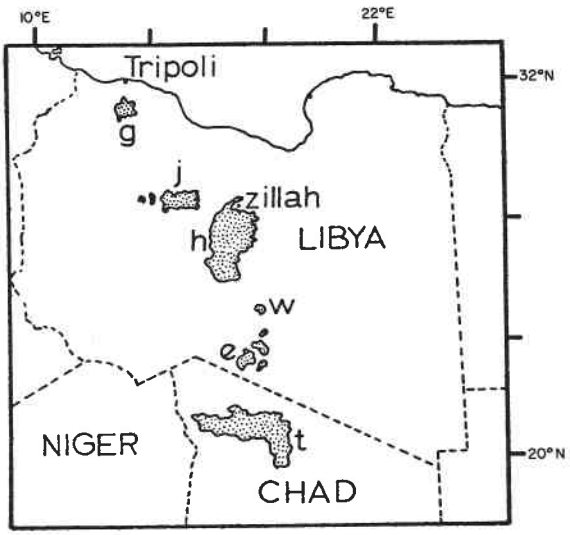


FIG. 1. The geographic distribution of the Cenozoic volcanic areas described in this series of papers.
g: Garian Volcanic area (Paper III)
d: Jebel Soda (this Paper, II)
h: Haruj Assuad (Paper I)
w: Waw en Namus
e: Eghei (Paper IV)
t: Tibesti (Paper IV)

Natural Remanent Magnetism of the Jebel Soda Volcanics

The natural remanent magnetization (NRM) of each specimen was measured using either a Schonstedt SSM-1a spinner magnetometer or a parastatic magnetometer. The latter employs feedback control of the displacement of the light spot. Systematic alternating field demagnetization was carried out. In most cases the stable component of magnetization dominated the N.R.M. Unit mean cleaned directions were obtained as described in Paper I. These directions are listed in Tables 2 and 3. Note that while a Fisherian α_{95} angle is listed to give a clear impression of the close grouping of within-unit cleaned directions, it is not implied that a group of 2 to 4 directions can be Fisherian.

The main point of interest concerning section A is the very close grouping of directions about the mean at 011°, +30°, with net variation in D and I of 5° and 7° respectively (Table 2). This small range in direction could have resulted from either extrusion during a period of very low secular variation or very rapid extrusion during a period characterized by more normal secular variation. The thin nature of the flows, and the lack of evidence for the weathering of one flow before it was covered by the succeeding flow, support the rapid extrusion explanation as being the correct one.

N.R.M. directions for the flows of section B show both N and R polarities and short sequences of closely grouped values as in section A together with a number of quite distinct directions (Table 3). This situation is consistent with the field evidence for periods of both rapid and slower extrusion rate. For example, flows B005 and 6 are thin flows without evidence of an intervening time break whereas the top of flow B011 is clearly reddened and 1.5 m red earth separates flows B014 and B015.

The first stage in attempting to obtain a best estimate of the pole for the dipole field direction during the period of volcanic activity is to weight the data to allow for the apparent large variations in extrusion rate and thus in the magnetic field sampling rate. The concept of the temporally independent direction is valuable here (Wilson 1970*a*, Paper I). The aim is to obtain the minimum number of temporally independent records of the geomagnetic field during extrusion of the flows. This is achieved by assuming that where directions of magnetization for successive flows differ by less than the respective α_{95} error

TABLE 1. Jebel soda volcanic area, geological nature and location of sections

Section	Flows	Geological nature and location
A	A10A001-A045 (A001 the youngest unit)	A section of 45 thin usually vesicular basalt flows frequently welded together forming a 70–100 m thick section at the head of a small dry stream system. The section is located from 1.0 to 1.5 km east of the Socna-Sebha highway at 48.5 km south of Socna. (Site is marked on a copy of aerial photograph V: 39/RAF/523 10 OCT 61 = 0912Z 61N 32000 ft REST'D number 0235)
B	A10B001-B017 (B001 the youngest unit)	A section of 17 thicker flows in an erosional scarp facing the Socna-Sebha highway. Several examples of red interflow material. Section thickness about 65 m (25 m covered by scree between flows B013 and B014). Senonian (Upper Cretaceous) sediments below B017. Section located 1.5 km to the east of the Socna-Sebha highway 34 km south of Socna and 7 km north of Bir Washka (Site is marked on a copy of aerial photograph V: 39/RAF/523 10 OCT 61 = 0912Z 61N 32000 ft REST'D number 0206)

TABLE 2. Jebel soda lavas, mean cleaned remanence directions for A section flows (A001 youngest)

Flow	D(°)	I(°)	k	α_{95}(°)	$n_{used}/n_{collected}$	Flow	D(°)	I(°)	k	α_{95}(°)	$n_{used}/n_{collected}$
A001	009	+32	414	2.4	4/4	24	012	+27	1111	1.5	4/4
2	005	+30	696	2.2	3/4	25	012	+28	1124	1.5	4/4
3	007	+29	494	3.2	2/4	26	006	+31	1855	1.2	4/4
4	005	+30	144	5.8	2/4	27	008	+31	287	2.9	4/4
5	012	+29	483	2.6	3/4	28	013	+28	608	2.0	4/4
6	011	+29	515	2.5	3/4	29	011	+33	768	2.1	3/4
7	010	+29	735	2.1	3/4	30	010	+29	1139	1.5	4/4
8	010	+29	1843	1.2	4/4	31	012	+30	446	2.3	4/4
9	009	+29	946	1.9	3/4	32	012	+31	728	1.8	4/4
10	006	+28	775	1.8	4/4	33	011	+31	923	1.6	4/4
11	011	+31	896	1.7	4/4	34	012	+31	863	1.7	4/4
12	006	+28	576	2.4	3/4	35	012	+31	317	2.8	4/4
13	005	+24	137	4.2	3/4	36	016	+35	118	6.5	2/2
14	No consistent direction				0/4	37	010	+34	1076	1.5	4/4
15	014	+30	1089	1.5	4/4	38	010	+40	1127	2.1	2/4
16	010	+35	575	2.1	4/4	39	015	+34	392	2.5	4/4
17	012	+34	396	2.5	4/4	40	015	+36	602	2.0	4/4
18	015	+32	268	3.0	4/4	41	Only one specimen, 012, +38 after 200 Oe				1/1
19	004	+31	664	2.2	3/4	42	010	+36	508	2.2	4/4
20	026	+28	207	3.4	4/4	43	Too soft to core in field				
21	013	+36	91	6.0	3/4	44	Too soft to core in field				
22	014	+29	204	4.0	3/4	45	015	+38	1419	1.3	4/4
23	008	+29	314	2.8	4/4						

limits the geomagnetic field has been sampled twice or more at the same point in time and space. In order to avoid redundancy, with bias in an average pole arising from this partial duplication of data, an average pole for the lava sequences is obtained by a two-tier operation, with averages for short successions representing repeated sampling at the same time being given unit weight in obtaining the overall average pole. This approach will suppresss evidence for long periods of low secular variation. However, geological evidence, such as the indication of a rapid extrusion rate for section A (and the intention of obtaining plate motion rather than details of the geomagnetic field) suggest that our approach is preferable. The 11 temporally independent paleomagnetic directions are listed in Table 3 and illustrated in Fig. 2. Several features of the distribution of directions are noteworthy. These are the azimuthal spread and the considerable offset of both N and R groups from the dipole field directions for the area. This offset is toward shallow inclinations, and is sufficiently large that directions have an equatorial aspect, with both shallow upwards and downwards inclinations characterizing the N group, although

TABLE 3. Jebel soda lavas, temporally independent paleomagnetic directions and pole positions

Section	Unit numbers	D(°)	I(°)	k	α_{95}(°)	λ(°N)	ϕ(°E)	δ_p(°)	δ_m(°)	$n_{used}/n_{collected}$	
A section	A001 to 45	011	+30	40	3.5	74	156	2.1	3.8	— See Table 2	
B section	B001	021	+05	218	3.8	57	154	1.9	3.8		
	B002 to 6	007	+12	481	2.9	66	177	1.5	2.9	—	003, +15; 008, +09; 007, +11; 012, +12; 008, +11
	B007 to 10	003	−01	837	2.8	60	188	1.4	2.8	—	005, −02; 8:Normal polarity 005, −02; 000, −01.
	B011	358	−09	280	3.3	57	199	1.7	3.4	4/4	
	B012	014	−33	121	8.4	42	178	5.4	9.5	2/2	
	B013	001	−24	179	4.3	49	195	2.4	4.6	3/4	
	B014	179	−42	490	2.6	85	209	1.9	3.2	3/4	
	B015	177	−14	668	2.2	68	203	1.2	2.3	3/4	
	B016	176	−21	194	3.6	72	208	2.0	3.7	4/4	
	B017	181	−13	308	2.8	68	194	1.5	2.9	4/4	

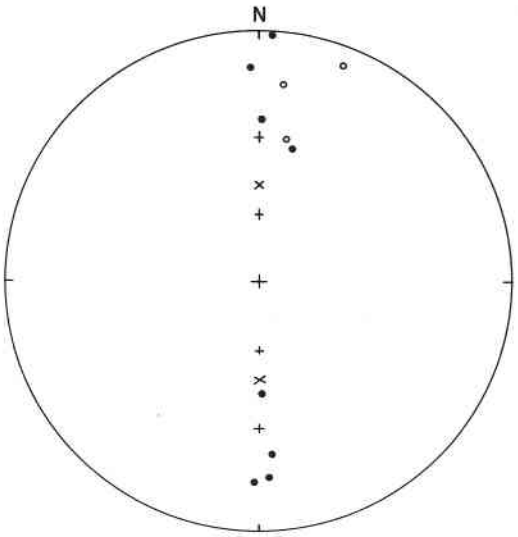


FIG. 2. Equal angle stereographic projection showing the temporally independent directions of cleaned NRM for the Jebel Soda basalts. Open circles: downwards inclinations. Full circles: upwards inclinations. Crosses: normal and reverse directions of magnetization for a centered axial dipole geomagnetic field source.

the site latitude is 28°N. The data of Schult and Soffel (1973) show similar features.

The problem from the plate motion point of view is to try to isolate from the data a best estimate of the dipole field configuration during the eruption of the lavas. Several guidelines exist but it is not certain that they are adequate. Thus flows B012 and B013 lie between fairly well defined R and N sequences and may have been magnetized during an intervening polarity transition of the field. If this is the case they can be omitted on the grounds that the dipole field is missing or at least, small compared with the non-dipole field during the polarity transition. Against this argument we note that these flows have azimuths typical of the normal directions, and in this respect are continuous with the overlying N sequence. Again, Wilson *et al.* (1972) have shown, for a large collection of Icelandic Neogene lavas, that the stable N and R states of the field are fairly sharply limited, in terms of equivalent pole positions, to latitudes in excess of 50°. Application of this criterion to the data of this paper again results in the rejection of B012 and B013, together with the data of Site 27 of the Schult and Soffel (1973) data. Even with this criterion applied there remain doubts as to the prospect of obtaining a representative dipole direction from the combined data. The combination of azimuthal spread of directions and large offset suggest that eruptive activity in the Jebel Soda bracketed a reversal of the geomagnetic field but had net duration too short to average out perturbations due to secular variation. Alternatively, the properties of the distributions of directions can be explained in terms of the existence of a rather permanent nondipole field in the Jebel Soda area during the period of activity. Simple vector averaging will not be sufficient to isolate the dipole component in the presence of such nondipole fields, which will produce a net bias in direction and pole distributions. With these uncertainties in mind we have computed a number of alternative average pole positions (Table 4 and Fig. 3). In the terms of conventional interpretation pole 7, at 73°N, 195°E, k: 26, A_{95}: 5.6°, is preferred. We note that every pole in the set differs significantly from the geographic pole, the range being from 12° to 25°, with the preferred pole deviating from the geographic pole by 17°. These deviations are probably too large to be explained by absolute plate motion. A number of other explanations, such as dipole offset and nonrepresentative sampling of the field may be appealed to as contributory effects. However, since a general reassessment of the methods of averaging paleomagnetic data, plus a consideration of the geomagnetic field in the North African area *as a whole* during the Cenozoic, need to be considered, we prefer to defer a full discussion to Paper IV of this series.

The pole positions for the temporally independent paleomagnetic directions may be used to estimate the secular variation for the area during the duration of volcanic activity. Following Doell (1970) and Cox (1969) and using the poles from the present study we obtain a value of $11.6°^{+5.2°}_{-2.7°}$ for the paleoangular standard deviation, S_F. This will be a maximum value if evidence for long periods of low secular variation has been suppressed by our method of obtaining temporally independent directions. This value of S_F is within the range of values predicted for several models of the geomagnetic field. However, as a result of the large uncertainty associated with the value, arising from the method of data reduction used, no one model can be preferred.

TABLE 4. Mean pole positions for the Jebel soda volcanic area

	N	Latitude (°N)	Longitude (°E)	k	A_{95}
For the data set of this paper					
(1) Using all 11 temporally independent poles	11	65	184	32	7.5
(2) Using 9 of the 11 temporally independent poles (B012/3 omitted)	9	69	184	47	6.8
For the data set of Schult and Soffel (1973)					
(3) Using all 16 sites	16	73	211	17	8.6
(4) Using 15 of the 16 sites (Sites 27/28/29/31 omitted)	15	75	205	22	7.7
(5) Using 12 of the 16 sites (Sites 27/28/29/31 omitted)	12	78	192	70	4.8
For the combined data of this paper and Schult and Soffel (1973)					
(6) Using all 27 available poles	27	70	197	19	6.2
(7) Using 24 of the 27 available poles (B012/3 and Site 27 omitted)	24	73	195	26	5.6

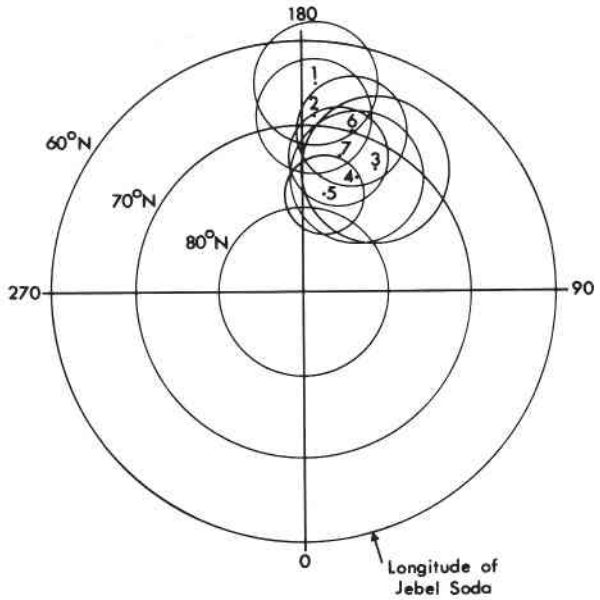


FIG. 3. Alternative locations of the average pole for the Jebel Soda lavas. For explanation of the numbers refer to Table 4.

Radiometric Dating of the Jebel Soda Volcanics

Following careful examination of thin sections of basaltic material from a number of flows, we selected four samples for age determination by the whole rock K–Ar method. Apart from some alteration of olivine (primarily to iddingsite), the samples appeared fresh and glass free.

A description of the dating equipment and the techniques used is given in Paper I. The results are summarized in Table 5. As a result of the relatively high potassium content and low atmospheric argon contaminations, well defined ages were obtained, with a small range in values from 10.5 m.y. to 12.3 m.y. The absolute ages indicate that the volcanic activity responsible for the A and B section flows was essentially contemporaneous and of less than 2 m.y. in overall duration. The age of the A and B section flows is close to the Middle-Upper Miocene boundary,

TABLE 5. Whole rock K–Ar ages of Jebel soda basalts

Unit	Core	^{40}Ar radiogenic (scc/g)	^{40}Ar radiogenic / ^{40}Ar total	%K_2O	Apparent age ± 2 standard deviations (m.y.)
A001	1	5.5×10^{-7}	60	1.4	11.6 ± 0.6
B012	1	6.1×10^{-7}	47	1.5	12.3 ± 0.5
B014	1	2.2×10^{-7}	22	0.62	10.5 ± 0.8
B017	1	4.4×10^{-7}	79	1.1	11.9 ± 0.5

Constants used: λ_ϵ: 0.585×10^{-10} y^{-1}
λ_β: 4.72×10^{-10} y^{-1}
^{40}K/K: 1.19×10^{-4} mole/mole

as defined in Berggren (1969). This is consistent with a Miocene (*sensu latu*) K–Ar age reported in Klitzsch (1970). The considerable differences in the ages of the exposed flows of the Jebel Soda and nearby Haruj Assuad volcanic areas, suggested by all authors, is now confirmed by absolute age data.

Summary

We find that the 10.5 to 12.3 m.y. Jebel Soda lavas provide excellent material for paleomagnetic study but that the well defined paleomagnetic directions are at least non-Fisherian in distribution and may also be biased towards the horizontal plane. The properties of these distributions seem likely to be original magnetizing field features as the lavas are little altered. A full discussion of these features of the direction distributions is reserved for a review paper, which will discuss paleomagnetic data and absolute ages from a number of North African Cenozoic volcanic areas.

Acknowledgments

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